Reliable and Invalid: Entropy Collapse Separates a Verifiable Reward from the Construct It Proxies

Anonymous Author(s)
Affiliation
Address
email

Abstract

Verifiable rewards are trusted because they are deterministic: a program either awards the points or it does not. We post-train Qwen2.5-7B-Instruct with GRPO on NYT Connections under a reward with a cheap structural component (well-formed output) and an expensive semantic component (correct groupings). On a strictly chronological test set of 162 puzzles (648 group slots), capability moves in four stages: the untrained Instruct model recovers 26 group slots, supervised fine-tuning 56, the step-50 GRPO policy 77, and the step-403 endpoint 4 — against a chance floor of 1.4 slots for a uniformly drawn valid partition. Step 50 improves on the untrained model by a paired $+0.315$ $[+0.204, +0.438]$ on the 0–4 scale; step 403 falls below it by -0.136 $[-0.191, -0.080]$, a deficit that reproduces across three seeds (all three point estimates inside that interval). The *training* reward mean records none of this: it reaches the reward's ceiling of 1.6 by step 125, with across-rollout variance zero on nearly every prompt from then on — the variance, not the mean, is the collapse's only in-sample trace — while the identical reward evaluated out of sample records the reversal: 0.246 at step 50, 0.266 at step 100 (separate serving session), 0.125 at the end, against the untrained model's 0.165. The semantic peak arrives by 5.7% of the run's eventual KL displacement; the remainder undoes it as policy entropy collapses and structural validity rises. The collapse survives disabling TRL's group-level reward normalization, and three LLM judges scoring the test generations blind all rank step 50 above the untrained model above the final policy — placing the *most* well-formed arm last. At 1.5B no arm is distinguishable from that chance floor, so the scale contrast reports what the reward did to the verifiable component and makes no claim about the construct. We read this as a construct-validity failure: the in-sample metric cannot locate the point at which a verifiable reward stops helping; the same reward, held out, brackets it.

1 Introduction

Post-training against verifiable rewards is attractive for a reason that sounds decisive: the reward is a program. It returns exactly what it was written to return at every point in policy space, and the argument that this buys immunity from reward hacking[1] rests on that reliability. This paper is about the gap reliability does not close: the gap between the reward and the construct it stands in for. We train on NYT Connections — sixteen words, partitioned into four groups of four — chosen because well-formedness and correctness are unusually separable: a response can be a perfectly valid partition of the board and completely wrong. Our reward pays for both, asymmetrically; two of its five components can be checked by reading the string, three require the answer key.

[1] We follow the common usage *reward hacking* throughout; Skalse et al. [2022], whose formalization we adopt, publish it under the name *reward gaming*.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

For fifty steps it works. Held-out groups recovered run $26 \rightarrow 56 \rightarrow 77$ across the untrained Instruct model, the supervised checkpoint GRPO is initialized from, and step 50 — the two stages contribute comparably (+26 to +30 supervised, +21 to +26 reinforcement learning; the SFT arm decodes to 56 or 52 slots depending on serving session, §3). Over the following 353 steps the same run falls to 4 slots, below the untrained model and near the 1.4-slot floor of a uniformly drawn valid partition, while the invalid-output rate falls by an order of magnitude and the training reward sits at 1.6, the maximum it can emit, with zero variance across the $K = 8$ rollouts on nearly every prompt. A practitioner watching that curve saw a solved task. The same reward function, evaluated on data the policy was not updated against, ranks all three checkpoints correctly. The failure is not in the metric; it is in the universal practice of reading it in sample.

Contributions. A fully verifiable, temporally leakage-controlled case study in which GRPO first improves and then destroys the target construct, ending below the untrained Instruct model across three seeds (§3); a checkpoint-level account placing the semantic peak at or before 5.7% of the run's KL displacement, co-occurring with entropy collapse and with the training reward at its ceiling — zero group-relative advantage on ~95% of prompts — from step 125 onward (§5); an ablation showing the collapse survives disabling TRL's group-level reward normalization (§6); a blind LLM-judge evaluation showing the reversal is legible from outside the reward (§7).

2 Task, pipeline, and the reward's asymmetry

Task and data. 1,078 dated NYT Connections puzzles (1,077 after validation), split strictly chronologically: 807 train (2023-06-12 to 2025-08-27), 108 validation, 162 test (2025-12-15 to 2026-05-29). Every test puzzle postdates every training puzzle; regressing groups-correct on puzzle date across the test window yields slopes indistinguishable from zero for both the untrained Instruct model (-0.158 groups/year $[-0.634, +0.292]$) and the trained policy ($+0.035$ $[-0.271, +0.329]$), so the split controls contamination; we cannot detect drift, though the intervals are wide (a 5.5-month window, slopes per year) and only large drift would show.

Pipeline. Qwen2.5-7B-Instruct (and 1.5B for the scale contrast), rank-16 LoRA. SFT for 3 epochs on the training split, then GRPO [Shao et al., 2024] from the SFT checkpoint: $K = 8$ sampled completions per puzzle at temperature 0.9, group-relative advantages, $\beta = 0.001$ against the SFT reference, learning rate 5×10^{-6}, one epoch (403 optimizer steps) at 7B.

The reward. Table 1 is the crux of the paper.

Table 1: The reward decomposition. Two components are verifiable from the emitted string alone; three require the answer key. The policy is trained on the sum.

Component	Value	Cost to check
Parses as four groups of four	$+0.1$	**cheap** — read the string
Correct groups	$+1.0 \times (\text{correct}/4)$	needs the answer key
All four groups correct	$+0.5$	needs the answer key
Group with exactly 3 correct words	$+0.05$	needs the answer key
Invalid output	-0.1	**cheap** — read the string

Maximum 1.6. The model can raise its score two ways: by getting better at the puzzle, or by getting better at emitting well-formed output. Only one of those is the construct. That asymmetry — cheap and expensive components, independently satisfiable — is the design property this paper studies.

3 The arc on held-out data

Table 2 reports the three checkpoints of the reported 7B run on the 162-puzzle test set, greedy decoding, all three arms served in a single vLLM session so every comparison is within-session.

Two reference points frame the table. A uniformly drawn valid partition recovers $4 \times 5775/2{,}627{,}625 = 0.0088$ groups per puzzle, or **1.4 slots** across this split; the step-403 endpoint

Table 2: 7B on the test split (162 puzzles, 648 group slots; one serving session; greedy). Means on the 0–4 scale are exact ratios of the count column. Bootstrap intervals are percentile, seed 0.

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward	Solves
Instruct (untrained)	0.1605 $[0.105, 0.222]$	26	6.8%	0.165	0/162
GRPO, step 50	**0.4753** $[0.364, 0.599]$	**77**	**16.7%**	**0.246**	2/162
GRPO, step 403	0.0247 $[0.000, 0.056]$	4	0.6%	0.125	0/162

sits at 2.8 times that floor — near enough that "below the untrained model" and "near a random valid partitioner" describe the same policy. Second, GRPO was initialized from the supervised checkpoint, not the untrained model. That checkpoint decodes to **56 slots** (0.346) in one serving session and 52 (0.321) in another, differing on 2 of 162 puzzles (Appendix B), so the stage accounting is $26 \to 52\text{–}56 \to 77 \to 4$: SFT contributes $+26$ to $+30$ and GRPO $+21$ to $+26$ at its peak — comparable contributions, and the data do not resolve which is larger. The SFT arm shares no serving session with Table 2's, so this is the one contrast we cannot pair; we report the range rather than either endpoint.

The method works, then reverses: $+0.315$ $[+0.204, +0.438]$ paired against the untrained model at step 50, -0.136 $[-0.191, -0.080]$ at step 403, both intervals excluding zero. Means in Table 2 are exact ratios of the count column; we report both. Two details locate the mechanism. Held-out mean reward reads 0.246 at step 50 and 0.125 at step 403, against base's 0.165, all in one session; in the later session of the plateau paragraph below, step 100 reads 0.2664 to a step-50 re-serve's 0.2525 — the highest held-out reward we measured: the reward peak, like the groups peak, is a plateau. The training reward meanwhile reaches its ceiling by step 125 (§5): the same reward ranks the checkpoints correctly out of sample and fails to on the training curve, and that contrast — not the reward's determinism — is this paper's subject. And the step-50 policy is worse than base on structure (16.7% invalid against 6.8%) but *not* worse than its actual initialization, the SFT arm at 22.2% — so what is specific to the phase after step 50 is the semantic collapse, not the structural climb, which was already more than half complete before the peak (§5).

The peak is a plateau, not a knife edge. Step 100 of the same run was additionally evaluated on test, in a single later serving session alongside a re-serve of step 50: 76/648 groups recovered (0.4691 $[0.3519, 0.6049]$) against step 50's 80/648 (0.4938 $[0.3827, 0.6235]$) *in that session*, paired difference -0.0247 $[-0.1481, +0.0927]$ ($n = 162$, seed-0 bootstrap) — statistically indistinguishable. The peak spans at least steps 50–100; the collapse occurs between steps 100 and 150, where the validation semantic score falls from 35/400 to 3/400 on the $n = 100$ temperature-0.9 entropy sweep (§5). The step-50 selection therefore does not rest on a one-column argmax.

Reproducibility across serving sessions. Base and the final checkpoint reproduce per-puzzle *exactly* across three independent vLLM sessions (0 of 162 differences each). The step-50 checkpoint does not: it differed on 2/162 puzzles across sessions (77 vs. 80 of 648) and on 6/162 completions between two greedy passes within one session. Mid-training policies sit near argmax ties that batched inference resolves inconsistently; converged ones have none left. That is evidence *for* the entropy account, and it is why every step-50 number in this paper carries its serving session. The published headline (77/648) is the original single-session evaluation; the plateau comparison above uses the later session's within-session pair.

Robustness of the endpoint. Three independent 7B seeds land at 4, 7 and 11 recovered slots of 648 (0.045 ± 0.022 on the 0–4 scale) — all below base's 26. Their LoRA update directions agree pairwise: cosine $+0.67$ to $+0.69$ between merged weight-delta directions, against $\mathrm{E}[|\cos|] = \sqrt{2/(\pi d)} \approx 10^{-5}$ for a random pair at $d \approx 6.5 \times 10^9$. Different random draws found the same attractor. Best-of-16 sampling does not rescue the converged policy, which is what a collapsed-entropy account predicts — a near-deterministic model has nothing new to sample; the single-seed pass@16 contrasts, read as illustrative only, are given with the per-arm values in Appendix B.

Conditioning on well-formed output widens the gap rather than explaining it, and we label the operation for what it is. A reviewer will ask whether the effect is an artifact of validity, since invalid generations score zero groups, and validity is what GRPO improved. Restricting to structurally valid generations at 7B, each arm's conditional computed within a single serving session (the SFT arm's is its 56-slot session): base **0.172** ($n = 151$ of 162), SFT **0.444** ($n = 126$), GRPO **0.025** ($n = 161$). This is a rescaling and we label it as one: each conditional value is that arm's unconditional mean divided by its validity rate, exactly, so the gap widens automatically whenever the low-scoring arm is the more valid one, which here it is by construction. What the rescaling licenses is therefore only the negative claim, which is the claim we need — the deficit is not an artifact of formatting, because it survives with formatting held fixed. Among answers that parse, the converged policy recovers 4 group slots where the untrained model recovers 26. (The 1.5B conditionals, on underlying counts of 1, 2 and 1, are given in Appendix C.)

One prediction of our own account fails, and we report it. If entropy collapse selectively destroyed board-specific inference, the loss should concentrate in the strata that need the most inference. It does not: across the 158 of the 162 test puzzles that carry a single stratum label, relative loss from the untrained model to GRPO runs 92% on `category` ($n = 69$), 67% on `cultural` ($n = 28$), 86% on `tag-fillin` ($n = 40$) and 100% on `wordplay` ($n = 21$, from a near-floor base of 0.048, i.e. a single recovered slot). That is closer to uniform destruction than to a gradient. We report it as a failed prediction of the selective-inference story rather than a refutation: at these stratum sizes the spread cannot separate uniform loss from a graded one.

4 The scale contrast at 1.5B

At 1.5B the instrument reads nothing. The untrained model recovers one group slot of 648, GRPO recovers one, and SFT — the best arm — recovers two, against the chance floor of 1.4 slots established in Section 3. No arm is distinguishable from that floor — SFT's two slots carry an exact binomial interval of $[0.0015, 0.0444]$ on the 0–4 scale (Appendix C), which contains it — so no claim about whether the construct was preserved is available at this scale. What is visible is that the reward did at 1.5B what it did at 7B on the component it can verify, driving invalid output from 32.1% to 2.5% and mean reward from 0.049 to 0.113 (full table in Appendix C). We report this as an observation rather than a contribution. The runs are epoch-asymmetric — 806 optimizer steps, two epochs, at 1.5B against 403 steps and one epoch at 7B — and we have no 1.5B checkpoint trajectory, so "the construct was untouched at 1.5B" would be a claim about an endpoint, which is exactly what this paper shows cannot settle such a question.

5 Mechanism: entropy collapse and the step-50 semantic peak

Figure 1 scores checkpoints of the reported run on the $n = 100$ temperature-0.9 *entropy sweep* (denominators of 400; full per-checkpoint values in Appendix A). The ablation's *checkpoint sweep* in §6 is $n = 108$, greedy, denominators of 432; the two are different measurements and are never mixed.

The semantic peak has a location; the structural climb does not. The semantic score rises from 0.0325 at the SFT initialization to 0.0950 at step 50 (13/400 to 38/400 recovered slots), then falls to 0.0075 by step 150 and never recovers. The structural score — the fraction of outputs that are a valid partition, not shown in Figure 1 — climbs across the same window and past it, with one reversal: 0.37 at the SFT initialization, 0.68 at step 50, 0.64 at 100, 0.95–0.96 from 150 on; of the 0.59 total structural gain, 0.31 is banked by step 50, concurrent with the semantic peak.

The trajectory is therefore not a trade with a turning point at which structure begins to be bought and semantics begins to be sold. It is a broadly monotone structural climb — one reversal, at step 100 — with a semantic hump on top of it, and only the hump has a location. One instrumentation gap qualifies the shape: no checkpoint exists between steps 100 and 150 — 81% of the eventual KL displacement (8.12 to 46.42 nats) falls in that window — so the descent's steepness and the turn's location are interpolated, not measured. The entropy account survives this reading unchanged — it predicts a policy that ends reliably well-formed and adaptively wrong, which is what the endpoint is — but it does not license the vocabulary of a trade and we do not use it.

Entropy collapse co-occurs with the loss. Policy entropy falls 13.4-fold between steps 50 and 150, the window in which the semantic score falls 12.7-fold; by step 150 the policy is effectively deterministic. A near-deterministic policy can be reliably well-formed — the format is the same for every board and can be satisfied without reading the puzzle. It cannot be adaptively correct, because

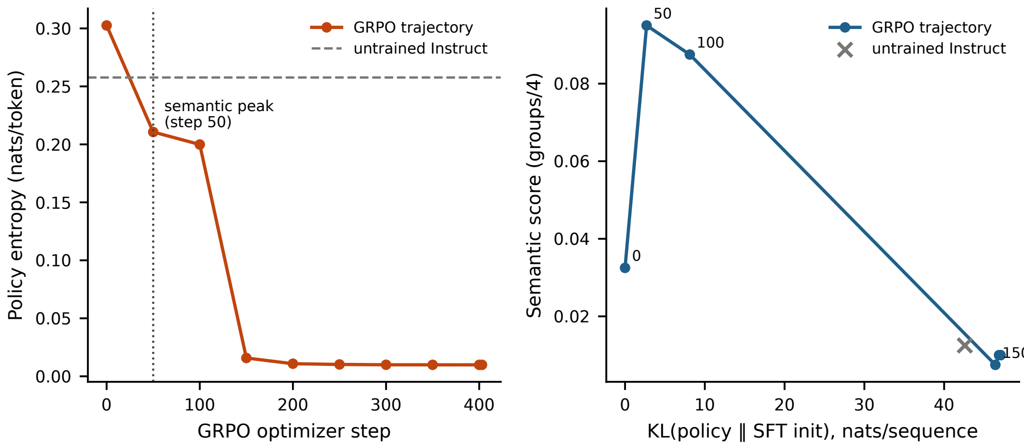

Figure 1: The mechanism, on the $n = 100$ temperature-0.9 entropy sweep (reported 7B run, seed 0). **Left:** policy entropy against optimizer step — a 30-fold collapse, with the cliff between steps 100 and 150. **Right:** semantic score against KL from the SFT initialization, the over-optimization curve: the peak arrives by 2.70 of an eventual 47.05 nats per sequence (5.7% of the run's displacement; step 50 is the earliest checkpoint retained), and the remaining 94.3% undoes it. The untrained Instruct model is plotted as an off-trajectory reference at 42.60 nats: KL from the initialization measures displacement, not damage (§5).

correctness requires responding to the specific sixteen words in front of it, which requires variation in the output. Collapse the variation and what remains is the component of the reward that does not depend on the input. An argument for plausibility, not a demonstrated mechanism: along this single path entropy, KL, structural validity and the zero-advantage fraction move together, and we never intervened on entropy (§10).

KL is the budget view, and it is a path length, not a damage scale. The step-50 checkpoint sits 2.70 nats from the SFT initialization and is the best policy this run produced on held-out data by a wide margin. The run spent 47.05. Stopping at 5.7% of the eventual KL displacement would have returned a model that recovers 77 group slots against base's 26; running to convergence returned one that recovers 4. Two qualifications keep this a path description, not a scale. Step 50 is the earliest non-initial checkpoint retained (nothing was measured at steps 10–40), so "at 5.7%" means *at or before* it. And KL from the initialization measures displacement, not damage: the untrained Instruct model sits 42.60 nats out on the same estimator — 90.5% of the run's displacement — while scoring 0.0125 on this sweep (Appendix A). Each estimate is taken under its own policy's samples, so base is far from SFT along a different direction, not a point on this trajectory; 5.7% dates the peak within this run and licenses no reading of KL as distance toward degeneracy. The sweep also compresses base's margin over the endpoint (5/400 vs. 4/400, where the greedy test instrument reads 26 vs. 4): temperature-0.9 sampling costs a high-entropy policy far more than a collapsed one — the instrument gap is the entropy account in miniature.

The training signal, read from the trainer's own logs. Figure 2 shows the paper's central contrast on one axis: the training reward first touches the reward function's ceiling of 1.6000 at step 125 and averages 1.5814 over the logged points from then on, with occasional excursions — 15 of 57 fall more than 0.02 below the ceiling, the deepest to 1.4300 at step 155 — and no trend. What the figure cannot show is TRL's own `frac_reward_zero_std` diagnostic: the fraction of prompt groups whose eight sampled rewards are identical, i.e., whose group-relative advantage is exactly zero. It rises from 0.0 at step 50 to 1.0 at step 125 and averages 0.946 over the remainder of training, so from step 125 onward the optimizer receives no learning signal on nearly every prompt. Read as a diagnostic: the in-sample mean records none of the reversal; the in-sample *variance* records most of it, bracketing the collapse between logged steps 100 and 150 with no held-out data. What the variance cannot supply is a sign — zero spread is what mastery and degeneracy both look like — so held-out evaluation still decides which. What drives the remaining 278 steps of movement while $\sim$95% of prompts carry

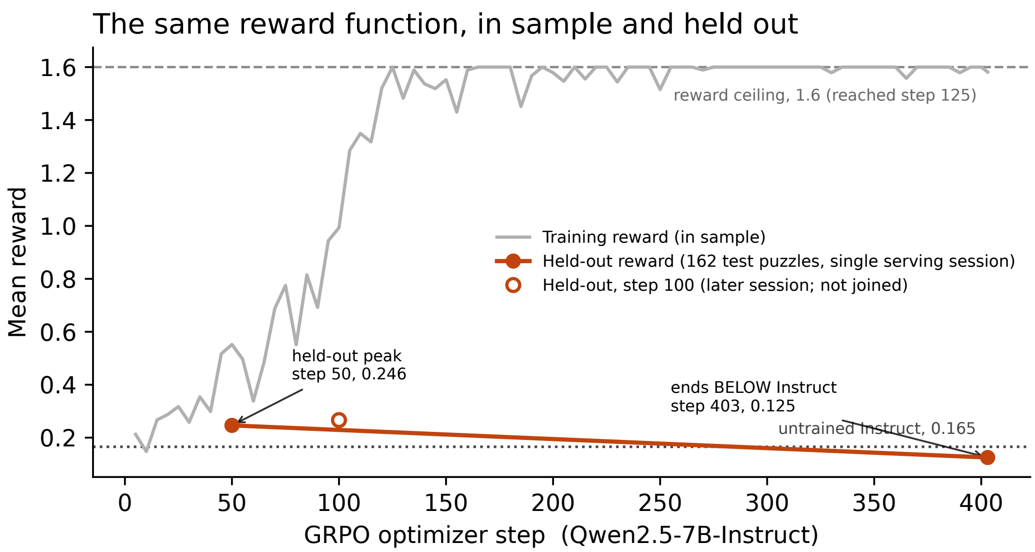

Figure 2: The same reward function, in sample and held out (reported 7B run, seed 0). The training reward (gray; TRL's 5-step interval averages from the run history) first touches the reward function's ceiling of 1.6 at step 125 and averages 1.5814 from then on, with occasional excursions and no trend. The held-out mean reward on the 162-puzzle test set (solid markers; single serving session) peaks at step 50 within its session and ends below the untrained Instruct model. The step-100 point (hollow marker) is from a later serving session — 0.2664 there, against a step-50 re-serve at 0.2525 — and is not joined to the curve.

zero advantage, we report as an open question; plausibly the residual prompts that still carry signal, compounded, but we have not measured that causal chain and do not assert it.

β was numerically inert, and the bracket is not an experiment. An earlier configuration at learning rate 10^{-6} and $\beta = 0.04$ froze the policy (KL ≈ 0.001 for 403 steps; greedy outputs byte-identical to SFT). The reported configuration, at 5×10^{-6} and $\beta = 0.001$, overshoots by roughly as much as the first undershoots. We state two qualifications immediately, because both weaken the pair as an experiment and neither weakens the observation. First, the two configurations differ in *two* variables at once, β by $40\times$ and the learning rate by $5\times$; a policy frozen at the lower learning rate is the expected consequence of the learning rate alone, so the pair bounds a region jointly and attributes nothing to either variable individually, and we do not claim to have measured inside a KL window. Second, $\beta = 0.001$ was numerically inert throughout the reported run: at step 50, $\beta \cdot \mathrm{KL} = 0.001 \times 2.70 = 0.0027$ reward units — eighteen times smaller than the one-away partial-credit term and 0.17% of the 1.6 maximum — and at convergence it reaches 0.047. The KL reference never restrained the policy at any point in the reported run, so this paper contains no evidence about what a KL penalty does when it is large enough to matter, and the intermediate-β sweep remains unrun.

6 The ablation: the collapse is not the normalization default

We inherited `scale_rewards="group"` from TRL — verified as the default in trl 1.10.0, whose `GRPOConfig.__post_init__` normalizes `False` to `"none"`. Adversarial reviewers of an earlier draft independently identified it as a confound: once the policy is near-deterministic, the within-group standard deviation that the advantage is divided by approaches zero, amplifying sampling noise into large updates, and that alone might produce the collapse, its timing, and the cross-seed agreement. Liu et al. [2025] argue against this normalization on independent grounds.

We reran the 7B pipeline identically — same SFT warm start, learning rate, β, K, temperature, LoRA rank, epoch count, seed — with reward scaling disabled, the resolved value read back off the constructed trainer configuration and logged. The signature is unchanged on the greedy

$n = 108$ checkpoint sweep: validation semantic score rises to 62/432 at step 50, falls to 35/432 by step 100 and 4/432 by step 150, while structural validity climbs 0.70 to 0.96. The validation-selected peak (step 50, frozen before any test puzzle was scored) recovers 87/648 on test (0.5370 $[0.4134, 0.6668]$, three solves), $+0.377$ $[+0.247, +0.506]$ paired against base. The final policy recovers 7/648 (0.0432 $[0.0123, 0.0804]$) — below base by -0.117 $[-0.179, -0.056]$ paired, and within $+0.019$ $[0.000, +0.043]$ of the original run's final policy.

The dynamics match as well: without reward scaling, policy entropy falls 0.2111 (step 50) $\rightarrow$ 0.1346 (100) $\rightarrow$ 0.0128 (150) and stays near 0.011 through step 403 (the original run's endpoint: 0.0099 nats/token), while KL from the SFT initialization rises 2.75 $\rightarrow$ 17.84 $\rightarrow$ 45.64 nats per sequence and plateaus near 46 (original: 47.05); on the $n = 100$ temperature-0.9 entropy sweep — a different measurement from the greedy $n = 108$ checkpoint sweep above — semantic score reads 35/400 at step 50, 17/400 at 100, then 2/400 from step 150 onward. The hump, the collapse, the entropy trajectory, and the below-base endpoint all survive with the normalization off. The phenomenon is not a library default.

7 The reversal is legible from outside the reward

The 162 test generations from base, step 50 and the final policy were scored blind — board words and completion only, no answer key — by three LLM judges (gpt-5-mini, gemini-2.5-flash, claude-sonnet-4-5) on a 1–10 scale, 162 items per arm per judge, 1458/1458 responses parsed. All three rank step 50 above base above the final policy, and all nine pairwise per-puzzle differences exclude zero under a paired bootstrap (seed 0); step-50 minus final is $+0.60$ $[+0.44, +0.78]$, $+3.25$ $[+2.81, +3.67]$ and $+1.62$ $[+1.33, +1.90]$ respectively. Every judge rates the collapsed policy below the untrained Instruct model despite its near-perfect output format. That ordering rules out the obvious confound: the final policy is the *most* well-formed arm on this split (0.6% invalid against 6.8% and 16.7%), so a shared preference for well-formed text would have placed it first, not last. Whatever the judges read, it is not the component the reward can still distinguish. The training reward cannot record the reversal; a blind judge can. Judges do err in correlated ways [Kohli, 2026], so we read this as three correlated readings of one external signal rather than three independent confirmations; per-item Pearson r across the three judge pairs is 0.456, 0.587 and 0.626 over all 486 items, lowest on the final policy's arm (0.200 to 0.433). The nine intervals are simultaneous and uncorrected; all exclude zero by margins no standard multiplicity correction would close.

8 Implications for evaluation practice

Distinguish in-sample from held-out proxy metrics before concluding a proxy has failed. Our reward does not fail as a ranking of these three checkpoints. Evaluated on held-out data it places step 50 above base above the final policy, which is the correct order. It fails only in the form practitioners actually consult it: as a training curve, in sample, where it reads perfect. The cheap remedy is not a better reward. It is the same reward, run on data the policy was not updated against — plus the trainer's own variance diagnostics, which bracket the collapse in sample though they cannot sign it (§5).

Report reward components separately, never the aggregate. This failure is invisible in the sum and obvious in the decomposition, and the decomposition is already computed. Component decomposition is a standard reward-model diagnostic; it is simply not applied to rewards that are deterministic, because determinism is taken to have settled the question.

Treat single-seed RL results as uninterpretable. Our own three seeds span 0.025 to 0.068 at the endpoint. A single seed cannot tell you where in that range you landed. This applies with equal force to results that come out well, including this paper's step-50 result: one seed, selected as the argmax of a nine-point validation grid on which step 100 scores 0.0875 to step 50's 0.0950 at $n = 100$ — inside one standard error. The grid preceded every test evaluation of the selected checkpoint (the ablation's peak was likewise frozen from validation first), so this is ordinary validation-based selection, and the step-100 test evaluation (§3) shows it rests on a plateau, not a lucky column.

9 Related work

Reward-model overoptimization. Gao et al. [2023] established this curve shape — gold-standard performance rising and then declining as KL from the initialization grows — for *learned* reward models, where the accepted explanation is Goodhart against an approximation, remediable in principle with a better reward model; Stiennon et al. [2020] report the same reversal. Our reward is a program: it approximates nothing, and there is nothing in it to game. The curve therefore requires only that a reward's cheap and expensive components be independently satisfiable — a property of its design, not its fit — and determinism is the exact property cited when practitioners argue that verifiable rewards escape this failure mode.

RL for reasoning and entropy collapse. Verifiable-reward RL underlies recent reasoning systems [DeepSeek-AI et al., 2025, Shao et al., 2024]. Yue et al. [2025] question whether RLVR extends capability beyond the pre-trained base model's pass@k ceiling. Our reference point differs (an instruction-tuned checkpoint, with RL initialized from a supervised checkpoint above it); here RLVR ends below *both* of its own antecedents on the construct. Cui et al. [2025] document entropy collapse as a central dynamic of RL for reasoning; we observe it co-occurring with the construct's destruction while the reward's cheap component still improves. Liu et al. [2025] critique group-level advantage normalization; our ablation (§6) shows the phenomenon here does not depend on it.

Reward hacking and its taxonomy. Skalse et al. [2022] formalize the failure mode (under the name *reward gaming*); Abahana [2026] classifies RLHF failure transitions by directional changes in learned reward and external judge scores, reporting that row-level diagnostics surface failures checkpoint averages hide. Two things differ here: our reward is rule-based, so the gap we report is not reward-model misgeneralization but a gap between a correct rule and the construct it stands in for; and our reversal is bracketed rather than flagged — the collapsed policy scores below the untrained Instruct model on held-out data, dating the turn instead of detecting it.

Validity in AI evaluation. Measurement-theoretic critiques of ML evaluation [Jacobs and Wallach, 2021, Freiesleben and Zezulka, 2025] distinguish a measurement from the construct it operationalizes; Chehbouni et al. [2025] apply the frame to LLM judges. Our contribution is a case where the measurement is maximally reliable — deterministic, verifiable, reproducible — and the validity gap does the damage anyway. Agarwal et al. [2021] motivate our statistical practice (paired bootstrap, counts alongside means, multi-seed reporting).

10 Limitations

One task, one reward design, chosen because it separates form from content unusually cleanly. The checkpoint trajectory is 7B seed 0 only (the only run Hub-synced during training) — so the scale-dependence claim rests on endpoints at both scales and a trajectory at one, and by our own argument an endpoint cannot rule out a traversed-and-returned peak at 1.5B (§4, which also notes the epoch asymmetry).

There is no human baseline. "Worse than the untrained Instruct model" is established; "bad in absolute terms" is not a claim we are entitled to make.

The headline checkpoint rests on one seed. Our three-seed evidence covers the converged endpoint, not the peak; its 0.025–0.068 spread is why we distrust single numbers elsewhere in this paper. The peak's height and location may both move across seeds — unmeasured — though the step-100 evaluation shows it is not a knife edge.

We report a single β alongside one that froze the policy, and no intermediate run; the two configurations differ in learning rate as well as β, and $\beta = 0.001$ contributed at most 0.047 reward units in the reported run. Entropy regularization and a larger KL penalty are untested recommendations, not findings.

The blind-judge evidence is three correlated scorers, not independent votes (§7). Training-reward saturation implies the policy solves its training prompts under sampling from step 125 on; whether SFT-stage memorization explains that is unmeasured — neither SFT nor the final policy has been scored on training puzzles — and open.

References

Zelalem Abahana. When RLHF fails: A mechanistic taxonomy of reward hacking, collapse, and evaluator gaming. *arXiv preprint arXiv:2606.03238*, 2026. Submitted 2 Jun 2026. Single-author preprint; no peer-reviewed version as of 24 Aug 2026.

Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron C. Courville, and Marc Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021. Outstanding Paper. arXiv:2108.13264.

Khaoula Chehbouni, Mohammed Haddou, Jackie Chi Kit Cheung, and Golnoosh Farnadi. Neither valid nor reliable? investigating the use of LLMs as judges. In *Advances in Neural Information Processing Systems 38 Main Conference (NeurIPS 2025)*, 2025. arXiv:2508.18076, 25 Aug 2025.

Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, Zhiyuan Liu, Hao Peng, Lei Bai, Wanli Ouyang, Yu Cheng, Bowen Zhou, and Ning Ding. The entropy mechanism of reinforcement learning for reasoning language models. *arXiv preprint arXiv:2505.22617*, 2025. Submitted 28 May 2025.

DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, et al. Deepseek-r1: Incentivizing reasoning capability in LLMs via reinforcement learning. *Nature*, 645:633–638, 2025. doi: 10.1038/s41586-025-09422-z. arXiv:2501.12948, 22 Jan 2025.

Timo Freiesleben and Sebastian Zezulka. The benchmarking epistemology: Construct validity for evaluating machine learning models. *arXiv preprint arXiv:2510.23191*, 2025. Submitted 27 Oct 2025.

Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 10835–10866. PMLR, 23–29 Jul 2023. arXiv:2210.10760, 19 Oct 2022.

Abigail Z. Jacobs and Hanna Wallach. Measurement and fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. Association for Computing Machinery, 2021. doi: 10.1145/3442188.3445901. arXiv:1912.05511.

Guneet Kohli. Nine judges, two effective votes: Correlated errors undermine LLM evaluation panels. *arXiv preprint arXiv:2605.29800*, 2026. Submitted 28 May 2026.

Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding R1-Zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025. Submitted 26 Mar 2025. Argues against group-level reward normalization.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. Introduces GRPO. Submitted 5 Feb 2024; v3 27 Apr 2024.

Joar Skalse, Nikolaus Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward gaming. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022. Published as "Reward Gaming"; circulated on arXiv as "Defining and Characterizing Reward Hacking", arXiv:2209.13085.

Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F. Christiano. Learning to summarize with human feedback. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020. Circulated on arXiv as "Learning to summarize from human feedback", arXiv:2009.01325.

Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems 38 Main Conference (NeurIPS 2025)*, 2025. Oral. arXiv:2504.13837, 18 Apr 2025.

A Per-checkpoint values for the entropy sweep

Table 3 tabulates all eleven rows of the sweep artifact: the trajectory plotted in Figure 1 (including the step-400 checkpoint, indistinguishable from step 403 at this precision), the structural column discussed in Section 5, and the off-trajectory untrained Instruct reference. All values are from the $n = 100$ temperature-0.9 entropy sweep of the reported 7B run (seed 0); the artifact is `results-analysis/entropy-kl-7b.json` in the released repository.

Table 3: The reported 7B run on the $n = 100$ temperature-0.9 entropy sweep. Structural = fraction of outputs that are a valid partition; semantic = mean correct groups / 4. KL is the per-sequence sample-based estimate against the SFT initialization, each row estimated under its own policy's samples; the untrained Instruct row is off the GRPO trajectory (§5).

Checkpoint	Policy entropy	KL from SFT init	Structural	Semantic
Instruct (untrained)	0.2577	42.60	0.69	0.0125
SFT init	0.3025	0.00	0.37	0.0325
Step 50	0.2107	2.70	0.68	**0.0950**
Step 100	0.1999	8.12	0.64	0.0875
Step 150	0.0158	46.42	0.95	0.0075
Step 200	0.0109	46.88	0.95	0.0100
Step 250	0.0102	46.96	0.96	0.0100
Step 300	0.0099	47.05	0.96	0.0100
Step 350	0.0099	47.05	0.96	0.0100
Step 400	0.0099	47.05	0.96	0.0100
Step 403	0.0099	47.05	0.96	0.0100

B 7B endpoints across three seeds

Table 4 tabulates the three-seed endpoint session discussed in Section 3; the 4 / 7 / 11 spread is the basis of the single-seed caveats in Sections 8 and 10. The pass@16 contrasts referenced there are differences of $+0.920$ $[0.778, 1.062]$ for SFT over GRPO and -0.370 $[-0.475, -0.259]$ for GRPO against the untrained Instruct model — single-seed, read as illustrative only.

Table 4: 7B endpoints across three seeds (separate serving session from Table 2; the SFT arm decodes to 0.346/56 in the Table 2 session's configuration and 0.321/52 here, differing on 2 of 162 puzzles; both GRPO anchor arms reproduce across sessions exactly).

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward
Instruct (untrained)	0.160	26	6.8% $[3.1, 11.1]$	0.165
SFT	0.321	52	22.2%	0.197
GRPO (3 seeds)	0.045 ± 0.022	4 / 7 / 11	1.2 ± 0.6%	0.132 ± 0.008

C 1.5B test-split table

Table 5 tabulates the 1.5B arms discussed in Section 4; every mean is the exact ratio of its count to 162.

Table 5: 1.5B on the test split (162 puzzles, 648 group slots). Groups-correct intervals are Clopper–Pearson exact binomial on the slot count of 648, scaled to 0–4 (a percentile bootstrap is degenerate at counts of 1–2); slots cluster within puzzles, so they are approximate. Invalid-rate intervals are percentile bootstrap over puzzles. Conditioned on structurally valid output (§3), the arms read: base 0.009 ($n = 110$ of 162), SFT 0.048 ($n = 42$), GRPO 0.006 ($n = 158$).

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward	Solves
Base	0.0062 $[0.0002, 0.0343]$	1	32.1% $[24.7, 38.9]$	0.049	0.0%
SFT	0.0123 $[0.0015, 0.0444]$	2	74.1% $[67.3, 80.2]$	-0.038	0.0%
GRPO	0.0062 $[0.0002, 0.0343]$	1	2.5% $[0.6, 5.6]$	0.113	0.0%